

Lecture 15: Causal inference

Economics 326 — Introduction to Econometrics II

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Motivation: causal questions

- Many important questions in economics are **causal**:
 - Does job training increase earnings?
 - Does a new drug improve health outcomes?
- The **fundamental challenge**: we can never observe the same individual both with and without treatment at the same time.
- This is modeled using the **potential outcomes framework**: each individual has two potential outcomes, one under treatment and one under control, but we only observe one of them.

Potential outcomes

- **Treatment status**: $D_i = 1$ if individual i receives treatment, $D_i = 0$ otherwise.
- For each individual i , define two **potential outcomes**:
 - $Y_i(1)$: outcome if individual i receives treatment ($D_i = 1$),
 - $Y_i(0)$: outcome if individual i does not receive treatment ($D_i = 0$).
- The **individual treatment effect** for person i is:

$$Y_i(1) - Y_i(0).$$

- Example: if Y_i is earnings and D_i indicates job training, then $Y_i(1) - Y_i(0)$ is the causal effect of training on earnings for person i .

The fundamental problem

- The **observed outcome** is:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

- If $D_i = 1$, we observe $Y_i(1)$ but not $Y_i(0)$.
- If $D_i = 0$, we observe $Y_i(0)$ but not $Y_i(1)$.
- We can never observe both potential outcomes for the same individual. This is the **fundamental problem of causal inference**.

Causal parameters of interest

- **Average Treatment Effect (ATE)**:

$$\text{ATE} = E[Y_i(1) - Y_i(0)].$$

- **Average Treatment Effect on the Treated (ATT):**

$$\text{ATT} = \text{E}[Y_i(1) - Y_i(0) \mid D_i = 1].$$

- **Conditional Average Treatment Effect (CATE):**

$$\text{CATE}(\mathbf{x}) = \text{E}[Y_i(1) - Y_i(0) \mid \mathbf{X}_i = \mathbf{x}].$$

- ATE averages over the entire population; ATT averages only over those who actually receive treatment; CATE conditions on observable characteristics $\mathbf{X}_i$.

Selection bias

- Since we cannot observe the counterfactual for each individual, a natural idea is to use the outcomes of **other people** as stand-ins: compare average outcomes of the treated group to average outcomes of the untreated group.
- Can we estimate ATE by simply comparing group averages, $\text{E}[Y_i \mid D_i = 1] - \text{E}[Y_i \mid D_i = 0]$?
- Comparing group averages seems natural, but it conflates the treatment effect with pre-existing differences between groups:

$$\begin{aligned} & \text{E}[Y_i \mid D_i = 1] - \text{E}[Y_i \mid D_i = 0] \\ &= \text{E}[Y_i(1) \mid D_i = 1] - \text{E}[Y_i(0) \mid D_i = 0]. \end{aligned}$$

- Add and subtract $\text{E}[Y_i(0) \mid D_i = 1]$:

$$\begin{aligned} &= \underbrace{\text{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} \\ &\quad + \underbrace{\text{E}[Y_i(0) \mid D_i = 1] - \text{E}[Y_i(0) \mid D_i = 0]}_{\text{Selection bias}}. \end{aligned}$$

- **Selection bias** arises when the treatment and control groups differ in their baseline outcomes $Y_i(0)$.
- E.g., people who choose to enroll in job training may differ systematically from those who do not.

Random assignment

- **Randomization** solves the selection problem: by randomly assigning treatment, we ensure the treated and control groups are comparable in expectation.
- Under **random assignment**, treatment D_i is independent of potential outcomes:

$$\text{E}[Y_i(0) \mid D_i = 1] = \text{E}[Y_i(0) \mid D_i = 0] = \text{E}[Y_i(0)].$$

- The selection bias vanishes, and the simple difference in means equals the ATE:

$$\begin{aligned} & \text{E}[Y_i \mid D_i = 1] - \text{E}[Y_i \mid D_i = 0] \\ &= \text{E}[Y_i(1) - Y_i(0)] = \text{ATE} = \text{ATT}. \end{aligned}$$

- Randomized experiments (like the National Supported Work program) achieve this by randomly assigning individuals to treatment and control groups.

Regression with a treatment dummy

- With random assignment, the ATE can be estimated by regressing Y_i on a treatment dummy:

$$Y_i = \alpha + \tau D_i + U_i,$$

where $E[U_i | D_i] = 0$ under randomization.

- The OLS estimate of τ equals the difference in sample means:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0,$$

where $\bar{Y}_1$ and $\bar{Y}_0$ are the sample averages for the treated and control groups.

Example: Lalonde data

- The **National Supported Work (NSW)** demonstration recruited disadvantaged workers (long-term unemployed, high-school dropouts, former drug users, ex-offenders).
- Among eligible applicants, some were **randomly assigned** to receive job training (treated group), and the rest formed the **experimental control group**. Both groups come from the same disadvantaged population.
- We use the `jtrain2` dataset from the `wooldridge` package, based on the Lalonde (1986) study:

```
library(wooldridge)
data(jtrain2)
set.seed(326)
jtrain2[sample(nrow(jtrain2), 10), c("train", "re78", "educ", "age", "black", "married")]
```

	train	re78	educ	age	black	married
357	0	3.34322	9	21	1	0
320	0	10.79860	8	18	0	0
108	1	1.95327	9	17	1	0
364	0	0.00000	11	39	1	0
256	0	0.00000	12	24	1	0
415	0	0.00000	8	23	1	0
222	0	2.01550	10	20	1	0
100	1	26.81760	9	31	0	0
145	1	2.48455	12	31	0	0
244	0	12.89840	9	22	1	0

- train**: 1 if randomly assigned to job training, 0 if assigned to control.
- re78**: real earnings in 1978 (thousands of dollars).

Estimating the ATE

- Since treatment was randomly assigned, the coefficient on **train** estimates the ATE:

```
summary(lm(re78 ~ train, data = jtrain2))$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.554802	0.4080460	11.162474	1.154113e-25
train	1.794343	0.6328536	2.835321	4.787524e-03

- The estimated ATE is approximately \$1,800 (**re78** is in thousands): on average, participation in the job training program increased 1978 earnings by about \$1,800.

Observational studies

- In many settings, randomization is not feasible or ethical. Treatment is determined by individual choices or institutional rules.
- Self-selection generates **selection bias**: workers who voluntarily enroll in job training may be more motivated; workers at firms offering 401(k) plans tend to have higher incomes.
- **Key idea**: if we can identify covariates X_i that explain why some individuals are treated and others are not, then **after controlling for** X_i , treatment may be as good as random.
- This is the **selection on observables** assumption: all confounding factors are captured by observable covariates.

Potential outcomes with a covariate

- The selection on observables assumption can be formalized as follows: after conditioning on X_i , treatment assignment D_i is as good as random.
- D_i and X_i can be correlated.
- Suppose the potential outcomes depend linearly on a covariate X_i :

$$Y_i(0) = \alpha_0 + \beta_0 X_i + U_i(0),$$

$$Y_i(1) = \alpha_1 + \beta_1 X_i + U_i(1),$$

- **Conditional mean independence assumption**:

$$E[U_i(0) \mid X_i, D_i] = 0 \quad \text{and} \quad E[U_i(1) \mid X_i, D_i] = 0.$$

- That is, after controlling for X_i , the residual terms $U_i(0)$ and $U_i(1)$ are uncorrelated with treatment D_i .
- The effect of X_i on the potential outcomes can differ between the treated and control groups: β_1 may not equal β_0 .

The ATE with a covariate

- Taking expectations of the potential outcomes:

$$E[Y_i(1)] = \alpha_1 + \beta_1 E[X_i],$$

$$E[Y_i(0)] = \alpha_0 + \beta_0 E[X_i].$$

- The ATE is:

$$\begin{aligned} \text{ATE} &= E[Y_i(1) - Y_i(0)] \\ &= (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)E[X_i]. \end{aligned}$$

- The ATE depends on the difference in intercepts **and** the difference in slopes, weighted by the population mean of X_i .

Two separate regressions

- One approach: estimate separate regressions for each group.
- **Control group** ($D_i = 0$): $Y_i = \alpha_0 + \beta_0 X_i + U_i(0)$.
- **Treatment group** ($D_i = 1$): $Y_i = \alpha_1 + \beta_1 X_i + U_i(1)$.
- The estimated ATE is:

$$\widehat{\text{ATE}} = (\hat{\alpha}_1 - \hat{\alpha}_0) + (\hat{\beta}_1 - \hat{\beta}_0) \bar{X},$$

where $\bar{X}$ is the overall sample mean of X_i .

Combined regression with interactions

- The observed outcome $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ can be written as a single regression. Expanding:

$$\begin{aligned} Y_i &= D_i \underbrace{(\alpha_1 + \beta_1 X_i + U_i(1))}_{Y_i(1)} + (1 - D_i) \underbrace{(\alpha_0 + \beta_0 X_i + U_i(0))}_{Y_i(0)} \\ &= \alpha_0(1 - D_i) + \alpha_1 D_i \\ &\quad + \beta_0 X_i(1 - D_i) + \beta_1 X_i D_i + (1 - D_i) U_i(0) + D_i U_i(1) \\ &= \alpha_0 + (\alpha_1 - \alpha_0) D_i \\ &\quad + \beta_0 X_i + (\beta_1 - \beta_0) X_i D_i + U_i, \end{aligned}$$

where $U_i = (1 - D_i) U_i(0) + D_i U_i(1)$.

- This is a regression of Y_i on D_i , X_i , and the interaction $X_i D_i$:

$$Y_i = \alpha_0 + \gamma D_i + \beta_0 X_i + \delta X_i D_i + U_i,$$

where $\gamma = \alpha_1 - \alpha_0$ and $\delta = \beta_1 - \beta_0$.

- The coefficient on D_i is $\gamma = \alpha_1 - \alpha_0$, which is **not** the ATE (unless $\beta_1 = \beta_0$).

The demeaning trick

- Recall that $\text{ATE} = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0) E[X_i]$, but the coefficient on D_i is only $\alpha_1 - \alpha_0$. To fix this, **add and subtract** $(\beta_1 - \beta_0) E[X_i] D_i$:

$$\begin{aligned} Y_i &= \alpha_0 + (\alpha_1 - \alpha_0) D_i + \beta_0 X_i + (\beta_1 - \beta_0) X_i D_i + U_i \\ &= \alpha_0 + [(\alpha_1 - \alpha_0) + (\beta_1 - \beta_0) E[X_i]] D_i \\ &\quad + \beta_0 X_i + (\beta_1 - \beta_0) X_i D_i - (\beta_1 - \beta_0) E[X_i] D_i + U_i \\ &= \alpha_0 + \underbrace{[(\alpha_1 - \alpha_0) + (\beta_1 - \beta_0) E[X_i]]}_{\text{ATE}} D_i \\ &\quad + \beta_0 X_i + (\beta_1 - \beta_0) (X_i - E[X_i]) D_i + U_i. \end{aligned}$$

- Defining $\tau = \text{ATE}$ and $\delta = \beta_1 - \beta_0$:

$$Y_i = \alpha_0 + \tau D_i + \beta_0 X_i + \delta (X_i - E[X_i]) D_i + U_i.$$

- The coefficient τ on D_i is exactly the **ATE**.

Estimating the ATE with covariates

- In practice, replace $E[X_i]$ with the sample mean $\bar{X}$ and run the regression:

$$Y_i = \hat{\alpha}_0 + \hat{\tau}D_i + \hat{\beta}_0X_i + \hat{\delta}(X_i - \bar{X})D_i + \text{residual}.$$

- The coefficient $\hat{\tau}$ on D_i directly estimates the ATE.
- This works because demeaning the interaction “absorbs” the $(\beta_1 - \beta_0)E[X_i]$ part of the ATE into the coefficient on D_i .

Why not regress Y_i on D_i and X_i ?

- A simpler regression omits the interaction:

$$Y_i = a + \tau D_i + bX_i + V_i.$$

- Recall $\delta = \beta_1 - \beta_0$ measures how the effect of X_i differs between the treatment and control groups.
- This is valid if $\beta_1 = \beta_0$ (the covariate has the same effect in both groups). Then $\delta = 0$, the interaction drops out, and the coefficient on D_i is the ATE.
- If $\beta_1 \neq \beta_0$ (the effect of X_i differs between groups), the omitted interaction creates bias: the interaction X_iD_i affects Y_i and is correlated with D_i , so the coefficient on D_i does not equal the ATE.

Example: separate regressions

- Estimate separate regressions of `re78` on `educ` for the treatment and control groups:

```
reg0 <- lm(re78 ~ educ, data = jtrain2, subset = (train == 0))
reg1 <- lm(re78 ~ educ, data = jtrain2, subset = (train == 1))
cbind(Control = coef(reg0), Treatment = coef(reg1))
```

	Control	Treatment
(Intercept)	3.80301658	-0.7821703
educ	0.07451936	0.6892860

- Compute the estimated ATE:

```
xbar <- mean(jtrain2$educ)
a0 <- coef(reg0)[1]; b0 <- coef(reg0)[2]
a1 <- coef(reg1)[1]; b1 <- coef(reg1)[2]
ATE_manual <- (a1 - a0) + (b1 - b0) * xbar
cat("Sample mean of educ:", round(xbar, 2), "\n")
```

Sample mean of educ: 10.2

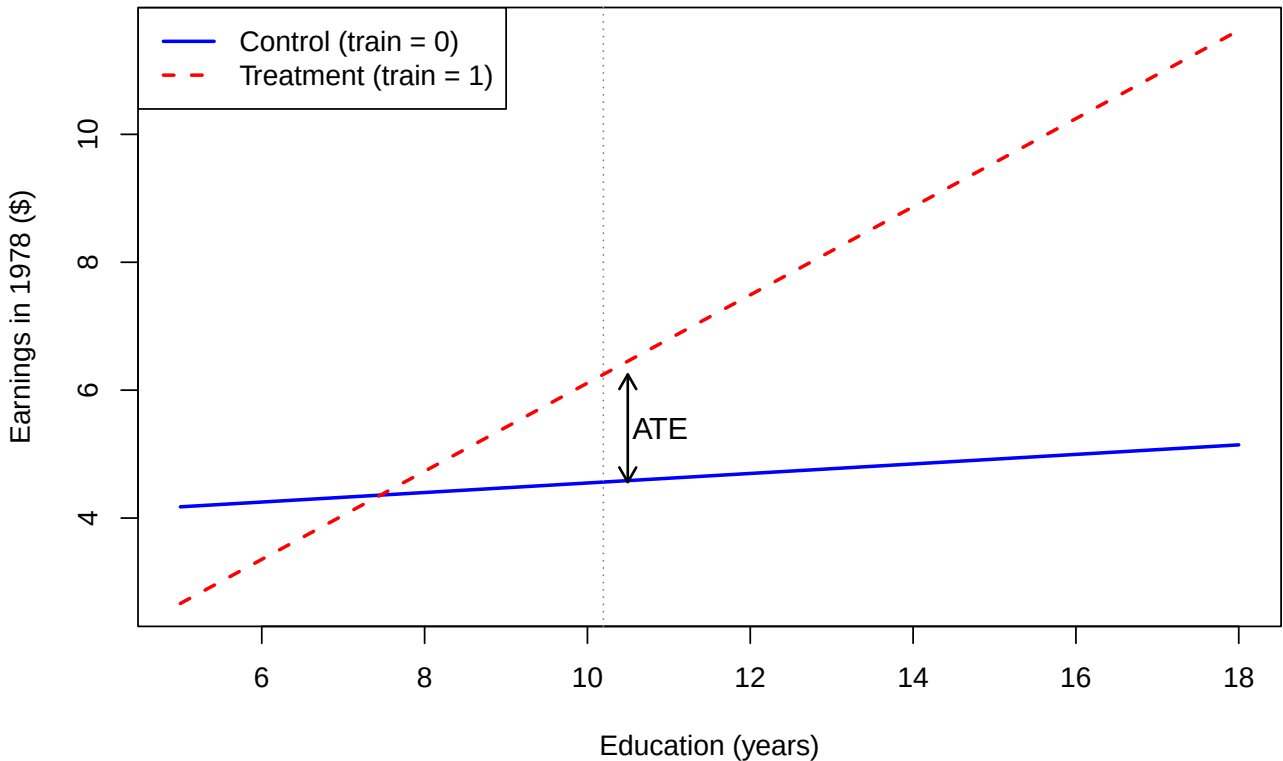
```
cat("Estimated ATE:", round(ATE_manual, 2), "\n")
```

Estimated ATE: 1.68

Example: regression lines

- The two regression lines, with a vertical line at $\bar{X}$ and the ATE marked as the gap:

Separate regression lines by treatment group



Example: demeaned regression

- Create the demeaned interaction and run the combined regression:

```
jtrain2$educ_dm <- jtrain2$educ - mean(jtrain2$educ)
reg_dm <- lm(re78 ~ train + educ + I(train * educ_dm), data = jtrain2)
summary(reg_dm)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.80301658	2.5689136	1.4803988	0.13948090
train	1.68266981	0.6299604	2.6710725	0.00784062
educ	0.07451936	0.2514521	0.2963561	0.76709766
I(train * educ_dm)	0.61476663	0.3472755	1.7702560	0.07737534

- The coefficient on `train` directly estimates the ATE, matching the result from the separate regressions approach.

Example: adding more covariates

- So far, we used only `educ` as the control. The dataset contains more pre-treatment characteristics: `age`, `black`, `married`.
- With random assignment, slopes are approximately equal across groups, so we can add controls directly without demeaned interactions:

```
reg_X <- lm(re78 ~ train + educ + age + black + married, data = jtrain2)
round(summary(reg_X)$coefficients, 1)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9	2.2	0.4	0.7
train	1.7	0.6	2.7	0.0
educ	0.4	0.2	2.4	0.0

age	0.1	0.0	1.2	0.2
black	-2.3	0.8	-2.7	0.0
married	0.2	0.8	0.2	0.9

Comparison: simple vs. controlled

- Compare the estimated ATE and its standard error across specifications, including one with demeaned interactions:

```
m1 <- lm(re78 ~ train, data = jtrain2)
m2 <- lm(re78 ~ train + educ + age + black + married, data = jtrain2)
covs <- c("educ", "age", "black", "married")
for (v in covs) jtrain2[[paste0(v, "_dm")]] <- jtrain2[[v]] - mean(jtrain2[[v]])
m3 <- lm(re78 ~ train + educ + age + black + married +
         I(train * educ_dm) + I(train * age_dm) +
         I(train * black_dm) + I(train * married_dm), data = jtrain2)
make_row <- function(m) c(Estimate = coef(m)["train"],
                          SE = summary(m)$coef["train", "Std. Error"])

tab <- rbind(
  "No controls"           = make_row(m1),
  "Controls, equal slopes" = make_row(m2),
  "Demeaned interactions" = make_row(m3)
)
round(tab, 2)
```

	Estimate.train	SE
No controls	1.79	0.63
Controls, equal slopes	1.68	0.63
Demeaned interactions	1.64	0.63

- All three estimates are nearly identical. The demeaned interactions barely matter because randomization ensures approximately equal slopes across groups.

Why do controls change little here?

- Under **random assignment**, D_i is independent of all covariates: the treatment and control groups are balanced in expectation.
- Because of balance, including X_i does not change the coefficient on D_i : there is no selection bias to remove.
- The NSW program recruited a narrow, disadvantaged population: long-term unemployed, high-school dropouts, former drug users, ex-offenders.
- Within this homogeneous group, covariates like **educ**, **age**, **black**, and **married** vary little and have limited predictive power for earnings.
- Both the point estimate and the standard error are nearly unchanged because the controls add almost no information beyond the treatment dummy.

Example: 401(k) eligibility and savings

- The `k401ksubs` dataset (from the `wooldridge` package) contains 9,275 households from the 1991 Survey of Income and Program Participation.

```
data(k401ksubs)
cat("n =", nrow(k401ksubs), "\n")
```

```
n = 9275
```

```
set.seed(326)
k401ksubs[sample(nrow(k401ksubs), 8), c("e401k", "nettfa", "inc", "age", "marr")]
```


	e401k	nettfa	inc	age	marr
1893	1	24.200	47.328	55	1
876	0	-1.000	13.260	45	0
927	0	0.000	17.571	37	1
6878	1	7.999	57.843	45	1
4196	0	8.000	13.800	25	0
1887	1	6.450	40.980	39	0
4355	0	29.700	22.410	34	0
7596	1	18.500	60.300	35	1

- **Treatment:** e401k = 1 if the worker's employer offers a 401(k) plan, 0 otherwise.
- **Outcome:** nettfa = net total financial assets (\$1000s).
- **Key covariate:** inc = family income (\$1000s).
- This is **observational**: eligibility depends on the employer, and higher-income workers tend to work at firms offering 401(k) plans.

The selection problem

- Group means by eligibility status:

```
grp <- split(k401ksubs, k401ksubs$e401k)
tab <- rbind(
  "Ineligible (e401k=0)" = colMeans(grp[["0"]][, c("nettfa", "inc", "age")]),
  "Eligible (e401k=1)"   = colMeans(grp[["1"]][, c("nettfa", "inc", "age")])
)
round(tab, 1)
```

	nettfa	inc	age
Ineligible (e401k=0)	11.7	34.1	40.8
Eligible (e401k=1)	30.5	47.3	41.5

- Eligible workers have much higher income and much higher savings. The raw gap in savings is **not** a treatment effect — it partly reflects income differences.

Separate regressions

- Estimate separate regressions of nettfa on inc for each group:

```
reg0 <- lm(nettfa ~ inc, data = k401ksubs, subset = (e401k == 0))
reg1 <- lm(nettfa ~ inc, data = k401ksubs, subset = (e401k == 1))
cbind(Ineligible = coef(reg0), Eligible = coef(reg1))
```

	Ineligible	Eligible
(Intercept)	-14.7353612	-25.105194
inc	0.7753202	1.176382

- Compute the estimated ATE from the two-regression formula:

```
xbar <- mean(k401ksubs$inc)
a0 <- coef(reg0)[1]; b0 <- coef(reg0)[2]
a1 <- coef(reg1)[1]; b1 <- coef(reg1)[2]
ATE_manual <- (a1 - a0) + (b1 - b0) * xbar
cat("Sample mean of inc:", round(xbar, 2), "\n")
```

Sample mean of inc: 39.25

```
cat("Estimated ATE:", round(ATE_manual, 2), "\n")
```

Estimated ATE: 5.37

Demeaned regression

- Create the demeaned interaction and run the combined regression:

```
k401ksubs$inc_dm <- k401ksubs$inc - mean(k401ksubs$inc)
reg_dm <- lm(nettfa ~ e401k + inc + I(e401k * inc_dm), data = k401ksubs)
summary(reg_dm)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-14.7353612	1.47212820	-10.009564	1.819941e-23
e401k	5.3737011	1.30596972	4.114721	3.910019e-05
inc	0.7753202	0.03654010	21.218339	1.320126e-97
I(e401k * inc_dm)	0.4010617	0.05286179	7.586988	3.589988e-14

- The coefficient on e401k directly estimates the ATE, matching the result from the separate regressions.

Comparison across specifications

- Four specifications for the effect of 401(k) eligibility on net financial assets:

```
m_none <- lm(nettfa ~ e401k, data = k401ksubs)
m_inc <- lm(nettfa ~ e401k + inc, data = k401ksubs)
m_int <- lm(nettfa ~ e401k + inc + I(e401k * inc), data = k401ksubs)

make_row <- function(m) c(Estimate = coef(m)["e401k"],
                          SE = summary(m)$coef["e401k", "Std. Error"])

tab <- rbind(
  "No controls" = make_row(m_none),
  "Income, no interaction" = make_row(m_inc),
  "Interaction (demeaned)" = make_row(reg_dm),
  "Interaction (not demeaned)" = make_row(m_int)
)
round(tab, 2)
```

	Estimate.e401k	SE
No controls	18.86	1.35
Income, no interaction	6.06	1.31
Interaction (demeaned)	5.37	1.31
Interaction (not demeaned)	-10.37	2.53

- No controls:** selection bias inflates the estimate — eligible workers earn more and save more regardless of 401(k) access.
- Income, no interaction:** removes much of the bias but constrains the model to equal slopes.
- Interaction (demeaned):** the coefficient on e401k is the ATE evaluated at mean income, showing 401(k) eligibility increases savings by about \$5,400.
- Interaction (not demeaned):** the coefficient on e401k is the **CATE** (conditional average treatment effect) evaluated at $\text{inc} = 0$:

$$\text{CATE}(x) = E[Y_i(1) - Y_i(0) \mid X_i = x] = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)x.$$

At $x = 0$ (zero income), this is an extrapolation far outside the data range — the **sign flips** to negative.

- Contrast with `jtrain2`: there, controls barely mattered because randomization already eliminated selection bias. Here, controls and the demeaning trick are essential.

Summary

- The **fundamental problem of causal inference** is that we never observe both potential outcomes $Y_i(1)$ and $Y_i(0)$ for the same individual.
- Comparing group averages $E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$ does not generally yield the ATE because of **selection bias**: treated and control groups may differ in their baseline characteristics.
- **Randomization** eliminates selection bias by making treatment independent of potential outcomes. The simple difference in means then equals the ATE, which can be estimated by regressing Y_i on a treatment dummy D_i .
- In **observational studies**, treatment is not randomly assigned. Under **selection on observables**, controlling for covariates X_i restores the “as good as random” property of treatment assignment.
- With covariates, $ATE = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)E[X_i]$ depends on both intercept and slope differences.
- The **demeaning trick** makes the ATE appear directly as the coefficient on D_i : interact D_i with $(X_i - \bar{X})$ instead of X_i . This absorbs the $(\beta_1 - \beta_0)E[X_i]$ component into the D_i coefficient.